Databases Management Systems

Not just philosophically, change indeed is the only constant. If you know a bit about programming, you'd also know how difficult it can be to design a storage structure only to realize later that it was imperfect and that improvements are needed. You might want to start using a database to solve the problem. But there was a time when they weren't around.

Databases can be said to have been developed in three stages. The first stage was the dawn of navigational databases in which the application would follow links or references to reach the final data. The second stage was the development of relational databases which stored data in rows and columns. Relational databases made more sense since they allowed the applications to search for data instead of following references and were better in terms of application growth. To this day, the relational model happens to be the most popular one. Oracle, Microsoft SQL Server, MySQL and PostgreSQL among others are all relational DBMS. The third generation of databases is called post-relational databases. The current generation of NoSQL DBMS such as MongoDB, HBase, Cassandra, Redis and CouchDB fall in this category.

Databases, especially relational ones, freed the programmers from having to take care of data storage and search so that they could focus on algorithms and features. Without databases, the time taken to develop applications would be at least 10 times greater. Database management systems also eliminated language dependencies as the data saved by a program written in one language (say C) could be read by a program written in another language (say Java). This decoupling